1. Să se calculeze :

$[1, 2, \dots, 100 \,;\, 15x^{100} + 2014]$

(1) $[x_1, \dots, x_{m+1} \,;\, \alpha \cdot f(x) + \beta \cdot g(x)] =$

$= \alpha \cdot [x_1, \dots, x_{m+1} \,;\, f] + \beta \cdot [x_1, \dots, x_{u+1} \,;\, g]$

(2) $[x_1, \dots, x_{u+1} \,;\, f] = \sum_{i=1}^{m+1} \frac{f(x_i)}{l'(x_i)}$, $f(x) = (x - x_1) \dots (x - x_{u+1})$

(3) $[x_1, \dots, x_{u+1} \,;\, x^k] = \begin{cases} 0, & k < m \\ 1, & k = m \\ S_1, & k = m+1 \end{cases}$

$[1, 2, \dots, 100 \,;\, 15x^{100} + 2014] =$

$= 15 \cdot [1, 2, \dots, 100 \,;\, x^{100}] + 2014 \cdot [1, \dots, 100 \,;\, x^0] =$

$\hookrightarrow (3) : \begin{cases} k = 100 \\ m = 99 \end{cases} \Rightarrow S_1$ $\hookrightarrow (3) : \begin{cases} k = 0 \\ n = 99 \end{cases} \Rightarrow 0$

$= 15 \cdot (1 + \dots + 100) + 2014 \cdot 0 =$

$= 15 \cdot 50 \cdot 101 = 15 \cdot 5050 = 75750$

2. Să se calculeze $[1, 2, 3, 4, 5; f]$ unde

$$f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)(x-6)\cdot x^{2014} + 2\sin(\pi x) + 2014$$

$$[1, \ldots, 5;\ (x-1)\ldots(x-6)\cdot x^{2014} + 2\sin(\pi x) + 2014] =$$

$$= \overbrace{[1, \ldots, 5;\ (x-1)\ldots(x-6)\cdot x^{2014}]}^{=0\ (2)} + [1, \ldots, 5;\ 2\sin(\pi x)] +$$

$$+ \overbrace{2014 \cdot [1, \ldots, 5;\ x^0]}^{=0\ (3)} =$$

$$= 2 \cdot \overbrace{[1, \ldots, 5;\ \sin(\pi x)]}^{=0\ (2)} = 0$$

3. Fie $x_0, \ldots, x_m$ noduri distincte în intervalul $[a, b]$ și $f : [a, b] \to \mathbb{R}$.

Să se arate că

$$[x_0, \ldots, x_m;\ (x-x_0)\ldots(x-x_k)\cdot f(x)] = [x_{k+1}, \ldots, x_n; f].$$

Pe baza relației (1) vom avea

$$\underbrace{\left[x_0, \ldots, x_k;\ \frac{(x-x_0)\ldots(x-x_k)\cdot f(x)}{(x-x_{k+1})\ldots(x-x_n)}\right]}_{=0\ (2)} + \left[x_{k+1}, \ldots, x_n;\ \frac{\cancel{(x-x_0)}\ldots\cancel{(x-x_k)}\cdot f}{\cancel{(x-x_0)}\ldots\cancel{(x-x_k)}}\right] =$$

$$= [x_{k+1}, \ldots, x_m; f] \qquad \text{Q.E.D.}$$

4. Demonstrați următoarea formulă pentru diferențe divizate

$$[x_0,\dots,x_m; f\cdot g] = \sum_{k=0}^{m} [x_0,\dots,x_k; f][x_k,\dots,x_n; g]$$

(T. Popoviciu)

$$[x_0, x_1, \dots, x_n; f] = \sum_{i=0}^{m} \frac{f(x_i)}{l'(x_i)}$$

$$[x_0, x_1, \dots, x_n; f] = \frac{[x_0, x_1, \dots, x_{n-1}; f] - [x_1, x_2, \dots, x_n; f]}{x_0 - x_m}$$

$$\sum_{k=0}^{m} \left(\sum_{i=0}^{k} \frac{f(x_i)}{l'(x_i)} \cdot \sum_{j=k}^{n} \frac{g(x_j)}{l'(x_j)} \right)$$

$$[x_0, \dots, x_n; f\cdot g] = [x_0, \dots, x_k; \frac{f(x)\cdot g(x)}{(x-x_{k+1})\dots(x-x_n)}]$$

$$+ [x_{k+1}, \dots, x_m; \frac{f(x)\cdot g(x)}{(x-x_0)\dots(x-x_k)}]$$

?

5. Fie $l_k(x)$, $k=\overline{0,m}$, $m \geq 1$, polinoame fundamentale Lagrange corespunzătoare nodurilor $x_k = \frac{k}{m}$, $k=\overline{0,m}$.

Să se calculeze $\frac{1}{m}\sum_{k=1}^{n} k \cdot l_k(x)$.

$$\boxed{L_m(P; x_0, \dots, x_m) \equiv P} \quad (1)$$

$\frac{1}{m}\sum_{k=1}^{m} k \cdot l_k(x)$ se notează cu $S(x)$

polin. fundam. pt. $x_k = \frac{k}{m}$, $k=\overline{0,m}$

$$S(x) = \frac{1}{m}\sum_{\underline{k=1}}^{n} k\, l_k(x) = \frac{1}{m}\sum_{\underline{k=0}}^{n} k\, l_k(x) = ($$ deoarece chiar dacă $k=0$, rezultatul nu va fi influențat)

$$= \sum_{k=0}^{m} \frac{k}{m} \cdot l_k(x) = (P_1 = x) = \sum_{k=0}^{m} P_1(x_k) \cdot l_k(x) =$$

$$\left(\begin{array}{l} P_1(x_k) = P_1\left(\frac{k}{m}\right) \\ P_1(x) = x, \text{ în cazul nostru } x = \frac{k}{n} \end{array} \right)$$

$$= L_n(f; x_0, \dots, x_m)(x) = \sum_{i=0}^{m} f(x_i) \cdot l_i(x) =$$

$$= L_n(P_1(x); x_0, \dots, x_n) = (1) = P_1(x) = x.$$

Să se calculeze

$$\frac{1}{n^2}\sum_{k=0}^{n} k^2 \cdot l_k(x) \quad , \quad m \geq 2$$

$$L_m(P; x_0, \dots, x_m) \equiv P \quad (1)$$

$\frac{1}{m^2}\sum_{k=0}^{n} k^2 \cdot l_k(x)$ se notează $S(x)$

$$S(x) = \frac{1}{m^2} \cdot \sum_{k=0}^{n} k^2 \cdot l_k(x) = \frac{1}{n^2}\sum_{k=0}^{n} k^2 \cdot l_k(x) =$$

$$= \sum_{k=0}^{n} \left(\frac{k}{m}\right)^2 \cdot l_k(x) = \left(P_2(x) = x^2\right) = \sum_{k=0}^{m} P_2(x_k) \cdot l_k(x) =$$

$$= L_m(P_2(x); x_0, \dots, x_n) = (1) = P_2(x) = x^2$$

Să se calculeze

$$\frac{1}{m^2}\sum_{k=0}^{n} (3k^2 + 4k + 5)\, l_k(\cancel{x})$$

$$L_m(P; x_0, \dots, x_m) \equiv P$$

$$\frac{1}{m^2}\sum_{k=0}^{n} (3k^2+4k+5) \cdot l_k(x) = \frac{1}{n^2}\left(\sum_{k=0}^{n} 3k^2 \cdot l_k(x) + \sum_{k=0}^{n} 4k \cdot l_k(x) + \right.$$

$$\sum_{k=0}^{n} 5 \cdot l_k(x) = \sum_{k=0}^{n} 3\left(\frac{k}{m}\right)^2 \cdot l_k(x) + \sum_{k=0}^{n} \frac{4}{m} \cdot \frac{k}{m} \cdot l_k(x) + \sum_{k=0}^{n} \frac{5}{m^2} \cdot l_k(x) =$$

$3x^2 + 4\frac{x}{n} + \frac{5}{n^2}$ (după alegerea fiecărui polinom)

6. Calculați următoarele diferențe divizate:

a) $[0, 1, \ldots, n ; \frac{x^{n+1}}{x+1}] = D_{a)}$

$$[x_1, \ldots, x_{n+1} ; x^k] = \begin{cases} 0, & k < n \\ 1, & k = n \\ S_1, & k = n+1 \end{cases}$$

$$\underbrace{[0, 1, \ldots n, -1 ; x^{n+1}]}_{= 0 \text{ (deoarece } k(=n+1) < n+2)} - D_{a)} + [-1 ; \frac{x^{n+1}}{x(x-1)\ldots(x-n)}] = 1$$

$$\Rightarrow D_{a)} = -1 + f(-1) = -1 + \frac{(-1)^{n+1}}{-1 \cdot (-1-1) \cdot (-1-2) \cdot (-1-n)} = -1 + \frac{(-1)^{n+1}}{-(n+1)!}$$

$$[x_i ; f] = f(x_i)$$

$$[x_i, x_k ; f] = \frac{f(x_k) - f(x_i)}{x_k - x_i}$$

b) $[0, 1, \ldots, n ; \frac{1}{x^2+1}] = D_{b)}$

$$\overbrace{[0, 1, \ldots, n, -i, i ; 1]}^{=0} - D_{b)} + [-i, i ; \frac{1}{x(x-1)\ldots(x-n)}] = 1$$

$$D_b = -1 + \frac{f(i) - f(-i)}{2i} \quad ?$$

$$c)\ \left[0,1,\dots,m;\ \frac{5x^{m+2}}{4x^2+12x+8}-4\cdot\frac{x^{m+1}}{x+2}\right]=D_{c)}$$

$$=\left[0,1,\dots,n;\ \frac{5x^{n+2}}{4(x+2)(x+1)}\right]-\left[0,1,\dots,n;\ \frac{4x^{n+1}}{x+2}\right]=$$

$$=\frac{5}{4}\cdot\underbrace{\left[0,1,\dots,m;\ \frac{x^{m+2}}{(x+2)(x+1)}\right]}_{:=D_1}-4\cdot\underbrace{\left[0,1,\dots,m;\ \frac{x^{m+1}}{x+2}\right]}_{:=D_2}=$$

$$D_1)\quad \underbrace{\left[0,1,\dots,m,-1,-2;\ x^{m+2}\right]}_{=0}-D_1+\left[-1;-2;\ \frac{x^{n+2}}{x(x-1)\dots(x-n)}\right]=1$$

$$\Rightarrow D_1=-1+\frac{f(-1)-f(-2)}{-1-(-2)}=-1+\frac{\frac{(-1)^{m+2}}{(-1)(-2)\dots(-1-n)}-\frac{(-2)^{n+2}}{(-2)(-3)\dots(-2-n)}}{1}=$$

$$=-1+\frac{(-1)^{m+2}}{-(n+1)!}-\frac{(-2)^{n+2}}{-(n+2)!}$$

$$D_2)\quad \underbrace{\left[0,1,\dots,m,-2;\ x^{n+1}\right]}_{=0}-D_2+\left[-2;\ \frac{x^{m+1}}{x(x-1)\dots(x-n)}\right]=1$$

$$\Rightarrow D_2=-1+f(-2)=-1+\frac{(-2)^{n+1}}{-(n+2)!}$$

$$D_{c)}=\frac{5}{4}\left(-1+\frac{(-1)^{m+2}}{-(m+1)!}-\frac{(-2)^{m+2}}{-(n+2)!}\right)-4\cdot\left(-1+\frac{(-2)^{m+1}}{-(n+2)!}\right)$$

d) $[1, \dots, m; x^{m+1}] = D_{d)}$

$x \cdot \Big| \; l(x) = (x-x_1)(x-x_2)\dots(x-x_{n+1}) = x^{n+1} - S_1 x^m + S_2 x^{n-1} - \dots$

$\Rightarrow x \cdot l(x) = x^{m+2} - S_1 x^{n+1} + S_2 x^n - \dots$

$[x_1, \dots, x_{n+1}; x \cdot l(x)] = 0 = D_{d)} - S_1^2 + S_2 \Rightarrow$

$D_{d)} = S_1^2 - S_2$

$[1, \dots, n, x^{n+1}] = (1+2+\dots+m)^2 - \sum_{\substack{i<j \\ i,j=1}}^{n} i \cdot j$

7. Se consideră următoarea problemă de interpolare: fiind date punctele distincte $x_0, \dots, x_n \in [a, b]$ şi $f : [a, b] \to \mathbb{R}$, să se determine o funcţie $G_m(f)$ de forma

$$G_m(f)(x) = \sum_{k=0}^{m} c_k \cdot e^{kx},$$

astfel încât

$$G_n(x_i) = f(x_i), \quad i = \overline{0,n}$$

să se arate că problema de interpolare de mai sus are soluţie unică.

Notăm $y := e^x$

$P_n(f)(y) = \sum_{k=0}^{n} c_k \cdot y^k = L_m(f; e^{x_0}, \dots, e^{x_m})(y)$

$y_i = e^{x_i} \in [e^a, e^b]$, y_i fiind distincte

Din problema Lagrange rezultă soluţia unică

8. Calculați următoarele diferențe divizate pe noduri duble:

a) $[x_0, x_0, \dots, x_m, x_m ; 5x^{2n+1} + x^{2n} + 3]$

$$[x_0, x_0, \dots, x_m, x_n ; f(x)] = \lim_{h \to 0} [x_0, x_0+h, \dots, x_m, x_{m+h} ; f(x)]$$

$$[x_0, x_0, \dots, x_m, x_n ; f(x)] = \sum_{i=0}^{m} \frac{f'(x_i)}{l'^2(x_i)} - \sum_{i=0}^{m} \frac{l''(x_i)}{l'^3(x_i)} \cdot f(x_i)$$

$$[x_0, x_0, x_1, x_1, \dots, x_m, x_n ; 5x^{2n+1} + x^{2n} + 3] =$$

$$= [x_0, x_0, \dots, x_n, x_n ; 5x^{2m+1}] + [x_0, x_0, \dots, x_m, x_n ; x^{2n}] + [x_0, x_0, \dots, x_n, x_n ; 3] =$$

$$= 5 \cdot [x_0, x_0, \dots, x_n, x_n ; x^{2m+1}] + [x_0, x_0, \dots, x_n, x_n ; x^{2n}] + 3 \cdot [x_0, x_0, \dots, x_n, x_n ; x^0] =$$

$$= 5 \cdot \lim_{h \to 0} [x_0, x_0+h, \dots, x_n, x_m+h ; x^{2n+1}] + \quad (n = 2n+2, \quad k = 2n+1) \Rightarrow 0)$$

$$+ \lim_{h \to 0} [x_0, x_0+h, \dots, x_n, x_n+h ; x^{2n}] + \quad (n = 2n+2, \ k = 2n \Rightarrow 0)$$

$$+ 3 \cdot \lim_{h \to 0} [x_0, x_0+h, \dots, x_n, x_n+h ; x^0] = \quad (n = 2n+2, \ k = 0 \Rightarrow 0)$$

$$= 0$$

b) $[x_0, x_0, \dots, x_n, x_n ; 4x^{2n+2}] = 4 \lim_{h \to 0} [x_0, x_0+h, \dots, x_n, x_n+h ; x^{2n+2}] =$

nr. elem $= 2n+2$, $k = 2n+2$

$4 \cdot 1 = 4$

9. Se consideră formula de cuadratură Gauss

$$\int_a^b w(x)\cdot f(x)dx = \sum_{k=0}^{n} A_k \cdot f(x_k) + R(f)$$

Fie $P_{m+1}(x)$ un polinom ortogonal relativ la ponderea $w:(a,b)\to\mathbb{R}$. Să se arate că:

$$A_k = \int_a^b w(x)\cdot \frac{P_{m+1}(x)}{(x-x_k)P'_{m+1}(x_k)}dx.$$

Formă de cuadratură Gauss $\Rightarrow x_0,\dots,x_m$ sunt rădăcinile lui $P_{n+1}(x)$

$$A_k = \int_a^b w(x)\cdot l_k(x)dx$$

$$P_{m+1}(x) = A_{m+1}(x-x_0)\dots(x-x_m) = A_{m+1}\cdot l(x)$$

$$l_k(x) = \frac{l(x)}{(x-x_k)\cdot l'(x_k)}$$

$$l_k(x) = \frac{P_{m+1}(x)}{(x-x_k)\cdot P'_{n+1}(x_k)} = \frac{\overset{1}{\cancel{A_{n+1}}}\cdot l(x)}{(x-x_k)\cdot \underset{1}{\cancel{A_{n+1}}}\cdot l'(x_k)}$$

10. Se consideră polinoamele lui Chebysev $T_{m+1}(x)$, polinoame ortogonale pe $(-1,1)$ relative la ponderea $w(x)=\frac{1}{\sqrt{1-x^2}}$,

$$T_{m+1}(x) = \cos((m+1)\arccos x)$$

a) Să se determine rădăcinile x_k, $k=\overline{0,n}$ ale lui $T_{n+1}(x)$.

$$T_{n+1}(x) = \cos((n+1)\arccos x)$$

$$\cos((n+1)\arccos x) = 0 \iff (n+1)\arccos x = \frac{(2k+1)\pi}{2} \iff$$

$$\iff \arccos x = \frac{(2k+1)\pi}{2n+2} \iff x = \cos\frac{(2k+1)\pi}{2n+2}$$

b) Să se demonstreze următoarea relație de recurență:

$$T_0(x) = 1,$$

$$T_1(x) = x,$$

$$T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x).$$

$$T_{n+1}(x) = \cos(n\cdot\arccos x + \arccos x) = \cos(n\cdot\arccos x)\cdot\cos($$

$$(\arccos x) - \sin(n\arccos x)\sin(\arccos x)$$

$$\Rightarrow T_{n+1}(x) = 2\cdot x\cdot T_n(x) - [\cos(n\cdot\arccos x)\cdot\cos(\arccos x) + \sin(n\arccos x)\cdot$$

$$\cdot\sin(\arccos x)] = 2\cdot x\cdot T_n(x) - \underbrace{\cos((n-1)\arccos x)}_{T_{n-1}(x)} =$$

$$= 2\cdot x\cdot T_n(x) - T_{n-1}(x)$$

c) Să se determine

$$\|T_{n+1}\|_w^2 := \langle T_{n+1}, T_{n+1}\rangle_w$$

$$\langle f, g\rangle_w = \int_a^b w(x)\cdot f(x)\cdot g(x)\,dx$$

$$\|f\|_w^2 = \langle f, f\rangle_w$$

$$\|T_{n+1}\|_w^2 = \langle T_{n+1}, T_{n+1}\rangle_w$$

$$\langle T_{n+1}, T_{n+1}\rangle_w = \int_{-1}^{1} w(x)\cdot T_{n+1}(x)\cdot T_{n+1}(x)\,dx$$

$$= \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\cdot \cos^2((n+1)\cdot \arccos(x))\,dx = 0 \text{, deoarece}$$

funcția de sub integrală este pară,

11. Să se determine nodurile și coeficienții formulelor de cuadratură de mai jos astfel încât acestea să aibă grad maxim de exactitate:

a) $$\int_{-1}^{1} f(x)dx = A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + R(f).$$

$w(x) \equiv 1$, $[a,b] = (-1,1)$

x_0, x_1, x_2 - rădăcinile lui $P_3(x)$

$$P_{n+1}(x) = \frac{d^{n+1}}{dx^{n+1}}\left[(x+1)^{n+1}(1-x)^{n+1}\right] = \frac{d^{n+1}}{dx^{n+1}}\left[(1-x^2)^{n+1}\right]$$

$$P_3(x) = \frac{d^3}{dx^3}\left[(1-x^2)^3\right] = \left[(1-x^2)^3\right]''' = \left[1-3x^2+3x^4-x^6\right]'''$$

$$= \left[-6x+12x^3-6x^5\right]'' = \left[-6+36x^2-30x^4\right]' =$$

$$= 72x-120x^3 = 24x(3-5x^2)$$

$$P_3(x) = 0 \Rightarrow x_0 = 0, \quad x_{1;2} = \pm\sqrt{\frac{3}{5}}$$

$$A_0 = \int_{-1}^{1} l_0(x)\,dx = \int_{-1}^{1} \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}\,dx = \frac{1}{(x_0-x_1)(x_0-x_2)} \int_{-1}^{1} x^2 - (x_1+x_2)\,x +$$

$$+ x_1 \cdot x_2\,dx = \frac{1}{(x_0-x_1)(x_0-x_2)} \cdot \frac{2}{3} \Rightarrow A_0 = \frac{5}{9}$$

$$A_1 = \int_{-1}^{1} l_1(x)\,dx = \int_{-1}^{1} \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}\,dx = \frac{1}{(x_1-x_0)(x_1-x_2)} \int_{-1}^{1} x^2 - (x_0+x_2)\,x + x_0 \cdot x_2\,dx =$$

$$= \frac{1}{+\sqrt{\frac{3}{5}} \cdot \left(-\sqrt{\frac{3}{5}}\right)} \cdot \int_{-1}^{1} x^2 - \frac{3}{5}\,dx = \frac{1}{-\frac{3}{5}} \cdot -2 \cdot \frac{3}{5} =$$

$$= -2$$

$$A_2 = \int_{-1}^{1} l_2(x)\,dx = \int_{-1}^{1} \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}\,dx = \frac{1}{(x_2-x_0)(x_2-x_1)} \cdot \int_{-1}^{1} x^2 - (x_0+x_1)\,x + x_0 x_1\,dx =$$

$$= \frac{1}{\sqrt{\frac{3}{5}} \cdot 2\sqrt{\frac{3}{5}}} \cdot \int_{-1}^{1} -\sqrt{\frac{3}{5}}\,x + 0\,dx =$$

$$= \frac{1}{2 \cdot \frac{3}{5}} \cdot (-2) \int_{0}^{1} \sqrt{\frac{3}{5}}\,x\,dx = \frac{1}{\frac{6}{5}} \cdot (-2) \sqrt{\frac{3}{5}}\frac{x}{2}\Big|_{0}^{1} = -\frac{5}{6} \cdot \sqrt{\frac{3}{5}}$$